A per-sample offset for scikit-learn logistic regression

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Abstract

scikit-learn's `LogisticRegression` cannot take a per-sample offset (a fixed term added to each example's log-odds), so a known reference probability can only enter as a feature. We add `offset=` to `fit` and every prediction method, and a package that uses a cross-validated fraction κ of the reference's log-odds as the offset. Against pristine scikit-learn 1.9.1 `LogisticRegressionCV` without the reference, test log-loss changes by -0.0529 (SE 0.0044) on HIGGS and -0.0882 (SE 0.0054) on Bank Marketing at 250 training rows, but not on the drifting Elec2 stream. Against the reference as a nearly unpenalised feature it ties, winning beyond 2 SE in 1 of 8 cells (HIGGS, $n = 1{,}000$: -0.0019, SE 0.0006). The one kept variant, a one-SE rule toward $\kappa = 1$, beat that feature under the same rule on fresh draws at $n = 100$ (-0.0103, SE 0.0050), not at $n = 150$, and never beat the reference alone. SEs are between disjoint repeats of paired held-out differences; fits match statsmodels and glum to 1.2×10^{-8}.

1 Problem

An offset is a known term added to the linear score of a generalised linear model (GLM) [McCullagh and Nelder, 1989]. The model learns weights for the features, but the offset keeps its coefficient fixed at one. R's `glm`, glmnet [Friedman et al., 2010], statsmodels [Seabold and Perktold, 2010] and glum [QuantCo, 2026] all accept it. scikit-learn [Pedregosa et al., 2011] does not, and the request to add it is still an open issue on the project's tracker [scikit-learn developers, 2022].

The case we care about is a strong reference: every example already has a probability from an older model, other data or an outside source. A ridge logistic model on a few hundred rows should then learn only the correction that the features add. Without an offset the reference's log-odds must go in as one more column, and the ridge shrinks the coefficient of the strongest input in the model. Scaling that column up makes its coefficient nearly free, which removes the shrinkage but also any pull toward one. The change came out of a private application, and Appendix C gives the one result we report from it. Everything else is measured on public benchmarks.

2 The change

We add an array `offset` of shape $(n,)$ to `fit`, `decision_function`, `predict`, `predict_proba` and `predict_log_proba`, in the same way as `sample_weight`. The model and its objective are

$$\eta_i = x_i^\top w + b + o_i, \qquad \min_{w,b} \; C \sum_{i=1}^{n} s_i \, \ell(y_i, \eta_i) + \tfrac{1}{2}\|w\|^2, \tag{1}$$

where η_i is the log-odds of example i, o_i its given (not fitted) offset, s_i its sample weight, C the inverse ridge strength and $\ell(y, \eta) = \log(1 + e^\eta) - y\eta$ the log-loss. The offset is data: it is never stored, and $o =$ `None` gives the current behaviour bit for bit. Our library diff is 232 added and 19 removed lines, with 318 lines of tests. One removed line fixes a bug the offset exposed: the Newton

line search evaluates $\eta(w + t\,d) = \eta(w) + t\,Xd$, but built Xd with the helper that adds o, so the offset counted twice per step and newton-cholesky stalled.

Our package `anchorlogit` builds the offset from a reference probability p_i, with $r_i = \mathrm{logit}(p_i)$:

$$o_i = \kappa\, r_i, \qquad \kappa \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}, \tag{2}$$

where κ is the fraction of the reference we trust, chosen jointly with C by 5-fold cross-validated log-loss. $\kappa = 1$ is the classical GLM offset and $\kappa = 0$ ignores the reference. For training rows the reference must be out of sample: if it was fitted on the same rows, `anchorlogit` cross-fits it (each row gets a prediction from a fit that did not see it) [Chernozhukov et al., 2018].

The recipe needs nothing specific to a domain: (i) For every example, training and new, get a reference probability p_i from a source that did not see that example's label, and clip it away from 0 and 1. (ii) Compute $r_i = \mathrm{logit}(p_i)$. (iii) Fit on the training rows with `offset=` κr for each (κ, C) on the grid, and keep the pair with the lowest cross-validated log-loss. (iv) At prediction time pass `offset=` κr for the new rows as well; the model returns the corrected log-odds. In the package, `AnchoredLogisticRegression(anchor=j, kappa="cv")` takes p as column `j` of X and does (ii) to (iv), with C tuned by any scikit-learn grid search; given an unfitted classifier as `anchor`, it also does (i) by cross-fitting.

A general form lets the coefficient on r move but pulls it toward κ, with an anchoring strength λ:

$$J(w, b, \beta) = C \sum_i s_i\, \ell\big(y_i,\ x_i^\top w + b + \beta r_i\big) + \tfrac{1}{2}\|w\|^2 + \tfrac{1}{2}\lambda(\beta - \kappa)^2. \tag{3}$$

$\lambda = \infty$ is the fixed offset of Equation 2 and $\lambda = 0$ is the reference as a free column. Lemma 1 shows that every λ in between needs nothing beyond `offset=`.

Lemma 1. *For $0 < \lambda < \infty$, (w^*, b^*, u^*) minimises the offset problem (1) with offset $o_i = \kappa r_i$ and one extra ridge column $r_i/\sqrt{\lambda}$ (weight u) if and only if $(w^*, b^*, \kappa + u^*/\sqrt{\lambda})$ minimises J.*

The proof is in Appendix A; the check in code matches the direct minimisation of J to 3.3×10^{-9} in η over 18 cases.

3 Setup

We use three public datasets from OpenML [Vanschoren et al., 2014], each with a different kind of reference. **HIGGS** [Baldi et al., 2014]: gradient boosting on the 21 low-level columns, fitted once on 60,000 disjoint rows, is the reference; the ridge sees all 28 columns, so the features partly re-express it. **Bank Marketing** [Moro et al., 2014]: gradient boosting on every column, fitted on a disjoint half, so the features add nothing (the expected null). **Elec2** [Harries, 1999]: a time-ordered stream; the reference is fitted on rows before each origin with an embargo (a gap of left-out rows) of 48 rows, so it ages, and the test set is the next week. Within a repeat every arm sees the same training rows, the same features and the same folds; only the use of r differs. HIGGS has 10 repeats, Bank 8 and Elec2 12; each repeat is a disjoint (train, test) draw with 1,500, 1,200 and 336 test rows.

Table 1: All arms share one cross-validation grid and the same folds, so the tuning budget is equal; a1 is the pristine upstream, a2 the trivial baseline. Platt scaling refits only the reference's slope and level.

Arm	Model	Arm	Model
a1	pristine `LogisticRegressionCV`, no reference	a5	offset, $\kappa = 1$ (classical GLM offset)
a2	reference alone (a2p: Platt [Platt, 1999])	a6	offset, (κ, C) by CV (ours)
a3	pristine, r as a standardised feature	a7	anchored, (κ, λ, C) by CV, $\lambda \in \{\infty, 10, 1, 0.1, 0\}$
a4	same, r times 100 (nearly free, $\lambda \approx 0$)	a9	a6, reference refit in-sample (leakage)
		a10	LightGBM [Ke et al., 2017] started at $\kappa_{\mathrm{cv}}\, r$

C runs over 11 values from 10^{-4} to 10. The metric is test log-loss in nats (natural-log units). For arm a against ours, repeat j of R gives the paired difference over its test rows T_j:

$$d_j = \frac{1}{|T_j|} \sum_{i \in T_j} \big[\ell(y_i, \eta_i^{\mathrm{ours}}) - \ell(y_i, \eta_i^{a})\big], \qquad \bar{d} = \frac{1}{R}\sum_{j=1}^{R} d_j, \qquad \mathrm{SE} = \frac{\mathrm{sd}(d_1, \ldots, d_R)}{\sqrt{R}}. \tag{4}$$

Negative $\bar{d}$ favours ours. Because the repeats are disjoint, this SE includes the noise of the training draw. We call it a win only when $|\bar{d}| > 2\,\mathrm{SE}$, and anything inside that is a tie. We fixed 3 checks before the runs: a6 beats a1 on most of the 3 datasets, a6 versus a3 and a4, and $\kappa_{\mathrm{cv}} < 1$ where the reference is off.

4 Results

In the main runs (Table 2) the offset beats the model without the reference in 6 of 8 cells. On HIGGS the gap falls from -0.0529 at $n = 250$ to -0.0276 at $n = 2{,}000$, and on Bank it is -0.0882 at $n = 250$.

The key comparison is the reference entered as a column. The penalised column a3 loses to the offset at small n (HIGGS $n = 250$: -0.0185, SE 0.0034; Bank: -0.0591, SE 0.0033), because the ridge shrinks its coefficient; the nearly free a4 does not. Against a4 the offset wins in 1 of 8 cells and loses in 0, and the anchored a7 also wins in 1. The single win is HIGGS at $n = 1{,}000$ (-0.0019, SE 0.0006), and in that cell a3 ties the offset (+0.0009, SE 0.0011). We therefore call the offset a tie with the free column.

Table 2: The offset ties the free column and beats the model without the reference on HIGGS and Bank, not on Elec2. Test log-loss per arm; $\bar{d}$ = a6 minus the arm with its between-repeat SE; bold is beyond 2 SE.

Data, n	a1	a2	a4	a5	a6	$\bar{d}$ vs a1	$\bar{d}$ vs a4	$\bar{d}$ vs a5
HIGGS 250	0.6839	0.6275	0.6321	0.6300	0.6309	$-$**0.0529** (0.0044)	$-$0.0012 (0.0020)	+0.0009 (0.0006)
HIGGS 500	0.6722	0.6275	0.6286	0.6287	0.6291	$-$**0.0431** (0.0046)	+0.0005 (0.0007)	+0.0004 (0.0004)
HIGGS 1,000	0.6536	0.6275	0.6278	0.6257	0.6259	$-$**0.0277** (0.0030)	$-$**0.0019** (0.0006)	+0.0002 (0.0002)
HIGGS 2,000	0.6461	0.6275	0.6187	0.6189	0.6186	$-$**0.0276** (0.0022)	$-$0.0002 (0.0002)	$-$0.0003 (0.0006)
Bank 250	0.2920	0.1991	0.2050	0.2051	0.2038	$-$**0.0882** (0.0054)	$-$0.0012 (0.0008)	$-$0.0013 (0.0009)
Bank 1,000	0.2548	0.1991	0.1984	0.2000	0.1986	$-$**0.0561** (0.0043)	+0.0002 (0.0005)	$-$0.0014 (0.0009)
Elec2 250	1.1382	0.9089	1.2465	1.3447	1.2175	+0.0793 (0.0465)	$-$0.0290 (0.0346)	$-$**0.1272** (0.0349)
Elec2 1,000	0.6864	0.9089	0.6940	0.8505	0.6926	+0.0062 (0.0098)	$-$0.0014 (0.0031)	$-$**0.1580** (0.0372)

In these runs the fraction κ mattered only where the reference was miscalibrated. On HIGGS cross-validation picks $\kappa = 1$ in 8/10 draws at $n = 250$, and a6 ties a5. On Elec2 the aged reference gets a median κ of 0.25 at $n = 1{,}000$, and a6 beats the full offset by -0.1580 (SE 0.0372); Figure 1 shows every comparison.

To find what would make the offset win, we tried 3 post-hoc variants and kept only the first. At $n = 100$ the CV minimum over κ was noisy, and κ_{cv} lost to $\kappa = 1$ (+0.0044, SE 0.0023). The one-SE rule [Hastie et al., 2009] takes, among all (κ, C) within one fold-SE of the best cross-validated loss, the largest κ and then the strongest ridge:

$$(\hat{\kappa}, \hat{C}) = \operatorname*{arg\,max}_{(\kappa,C)\in S} (\kappa, -C), \qquad S = \{(\kappa, C) : \mathrm{CV}(\kappa, C) \le \mathrm{CV}_{\min} + \mathrm{SE}_{\min}\}, \tag{5}$$

where the arg max is lexicographic (compare κ first, then C) and $\mathrm{SE}_{\min}$ is the fold SE at the CV minimum. To keep the comparison fair, the free column gets the same C rule as a control. Figure 1(d) shows the result and Table 5 in Appendix B lists it.

On the draws it was chosen on, the rule beats the free column under its own C rule at $n = 150$ (-0.0064, SE 0.0030), and at $n = 100$ it is just inside 2 SE (-0.0082, SE 0.0044). Because it was chosen after seeing those numbers, we reran it on 5 fresh HIGGS draws whose rows none of the variant runs used. There the order flips: a win at $n = 100$ (-0.0103, SE 0.0050) and a tie at $n = 150$ (-0.0021, SE 0.0016). It loses to the reference alone at every $n \le 250$, on the fresh draws too. Because it selects $\kappa = 1$ in every draw, this rule amounts to the classical offset with a conservative ridge, not a fractional one.

A reference built from all 28 HIGGS columns, which the features only re-express, gave a tie with the free column (+0.0003, SE 0.0027, $n = 100$) and a loss to the reference alone. A Bank reference trained only on younger clients, wrong in a group the features can see, is where the fraction pays: κ_{cv} has median 0.75 and beats $\kappa = 1$ by -0.0077 (SE 0.0012) at $n = 1{,}000$. It still ties the free column (-0.0004, SE 0.0003) and the Platt-recalibrated reference.

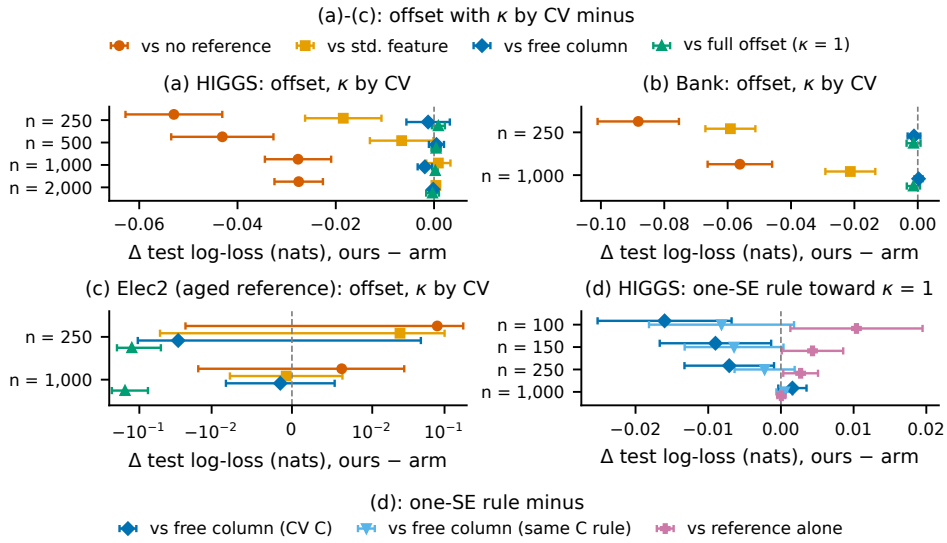


Figure 1: The offset beats no reference and the penalised column, and sits on zero against the free column. Paired test log-loss difference with 95% intervals from the between-repeat SE; (c) has a symmetric log axis.

Appendix B gives the ablations and equivalence checks. Calibration tells the same story: in Figure 2 the expected calibration error (ECE, mean gap between predicted and observed frequency) of the offset and of the free column are close.

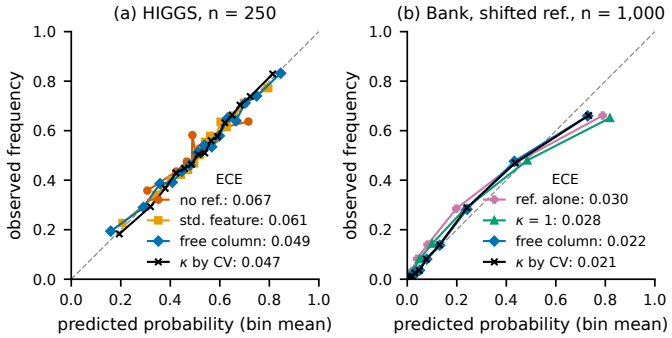


Figure 2: The offset is as well calibrated as the free column. Reliability on pooled test rows, 15 equal-mass bins; legend ECE is the mean over repeats.

5 Limits

The headline is a tie: the offset beats the nearly free column in 1 of 8 cells (1 under a two-stage bootstrap SE that resamples repeats, then test rows), so scaling one column gets most of the gain.

On Elec2 the offset does not beat the model without the reference (+0.0793, SE 0.0465); the Platt-recalibrated reference reaches 0.6347 against its 1.2175 at $n = 250$ (+0.5828, SE 0.2918, a tie only by the large SE), so $\kappa < 1$ is a coarse fix for a reference wrong in slope and level.

We ran 12 main arms and 3 variant arms over 7 result files. Of the 3 pre-registered checks only the win over a1 on Elec2 failed, and a6 tied a3 at $n \geq 1{,}000$. The 3 variants were post-hoc, and the kept one held on fresh draws at one of two sizes, by about 2 SE: weak evidence at best. ACS Income, $n > 2{,}000$ and more repeats were cut for budget. The method is binary only; we claim nothing for the Poisson, Gamma and Tweedie regressors beyond the patch's tests.

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A Proof of Lemma 1

Proof of Lemma 1. Given:

1. J as in (3);

2. the offset problem with the extra column, $K(w,b,u) = C\sum_i s_i\,\ell\big(y_i,\ x_i^\top w + b + u\,r_i/\sqrt{\lambda} + \kappa r_i\big) + \frac{1}{2}\|w\|^2 + \frac{1}{2}u^2$;

3. if ϕ is a bijection and $K(v) = J(\phi(v))$ for all v, then v^* minimises K if and only if $\phi(v^*)$ minimises J.

Choose: $\phi(w,b,u) = (w,\ b,\ \kappa + u/\sqrt{\lambda})$, so $\beta = \kappa + u/\sqrt{\lambda}$.

Need to prove: $J(w,b,\kappa + u/\sqrt{\lambda}) = K(w,b,u)$ for all (w,b,u). The ridge term $\frac{1}{2}\|w\|^2$ is the same in both. The score term:

$$\begin{aligned} \beta r_i &= \big(\kappa + u/\sqrt{\lambda}\big) r_i && \text{substitution of } \beta \\ &= \kappa r_i + u\,\big(r_i/\sqrt{\lambda}\big) && \text{distributive law} \end{aligned}$$

The anchoring term:

$$\begin{aligned} \tfrac{1}{2}\lambda(\beta-\kappa)^2 &= \tfrac{1}{2}\lambda\big(u/\sqrt{\lambda}\big)^2 && \text{substitution of } \beta \\ &= \tfrac{1}{2}\lambda\,u^2/\lambda && \text{power of a quotient} \\ &= \tfrac{1}{2}u^2 && \text{cancellation, } \lambda \neq 0 \end{aligned}$$

Substitution of both pieces into J, with the same ridge term, gives:

$$\begin{aligned} J(w,b,\kappa + u/\sqrt{\lambda}) &= C\sum_i s_i\,\ell\big(y_i,\ x_i^\top w + b + u\,r_i/\sqrt{\lambda} + \kappa r_i\big) && \text{both pieces} \\ &\quad + \tfrac{1}{2}\|w\|^2 + \tfrac{1}{2}u^2 \\ &= K(w,b,u) && \text{definition of } K \end{aligned}$$

Precondition: rule 3 is only valid if ϕ is a bijection, therefore its inverse must exist on all of $\mathbb{R}$:

$$\begin{aligned} \beta = \kappa + u/\sqrt{\lambda} &\iff \beta - \kappa = u/\sqrt{\lambda} && \text{subtract } \kappa \\ &\iff u = \sqrt{\lambda}\,(\beta - \kappa) && \text{multiply by } \sqrt{\lambda},\ 0 < \lambda < \infty \end{aligned}$$

Every $\beta \in \mathbb{R}$ has exactly one such u, and w and b map to themselves, so ϕ is a bijection.

Therefore, by rule 3: (w^*, b^*, u^*) minimises K if and only if $(w^*, b^*, \kappa + u^*/\sqrt{\lambda})$ minimises J. □

B Ablations and checks

Table 3 changes one part of the method at a time on HIGGS. Fixing $\kappa = 1$ (a5) or adding the anchoring strength (a7) changes nothing beyond 2 SE. Dropping from the ridge the 21 columns the reference was built from (a8) helps at $n \geq 500$: those columns only re-express the reference and cost variance. A reference refit on the training rows must be cross-fitted: refit in-sample it leaks the labels, and at $n = 250$ its log-loss is 1.1443 against 0.6844 cross-fitted. A nonlinear correction beats every linear arm once $n \geq 500$: LightGBM started at the same offset (a10) is better than a6 by +0.0387 (SE 0.0026) at $n = 1{,}000$.

Table 3: On HIGGS fixing $\kappa = 1$ or adding the anchoring changes nothing; dropping the reference's source columns, the source of the reference and the model class do. $\bar{d}$ = a6 minus the arm (SE), 10 repeats; positive means the arm is better; bold is beyond 2 SE.

n	a5: $\kappa = 1$	a7: anchored	a8: no source columns	a9: refit, cross-fitted	a9: refit, in-sample	a10: LightGBM
250	+0.0009 (0.0006)	−0.0020 (0.0021)	+0.0030 (0.0026)	**−0.0535** (0.0040)	**−0.5133** (0.0358)	**−0.0119** (0.0052)
500	+0.0004 (0.0004)	+0.0002 (0.0010)	**+0.0062** (0.0019)	**−0.0431** (0.0046)	**−0.3737** (0.0218)	**+0.0210** (0.0021)
1,000	+0.0002 (0.0002)	−0.0005 (0.0003)	**+0.0089** (0.0014)	**−0.0277** (0.0030)	**−0.1717** (0.0044)	**+0.0387** (0.0026)
2,000	−0.0003 (0.0006)	+0.0000 (0.0000)	**+0.0049** (0.0016)	**−0.0259** (0.0022)	**−0.0827** (0.0032)	**+0.0473** (0.0032)

The implementation itself is exact (Table 4). 61 new upstream tests pass on scikit-learn 1.9.1, and the package passes 295 tests with 0 failures.

Table 5 lists the one-SE rule of Section 4 against its controls.

Table 4: The patched fit matches the reference GLM implementations and stock scikit-learn to solver tolerance. Largest absolute difference over the cases.

Check	cases	coefficients	log-odds η
patched fit vs statsmodels GLM with offset	12	8.3×10^{-9}	3.6×10^{-8}
patched fit vs glum with offset	18	1.2×10^{-8}	4.4×10^{-8}
zero offset vs stock scikit-learn	9	1.9×10^{-8}	not stored
Lemma 1: offset + column vs direct minimisation of J	18	2.5×10^{-10}	3.3×10^{-9}

Table 5: The one-SE rule beats the free column at small n by about 2 SE, not stably across draws, and never beats the reference alone. $\bar{d}$ = rule minus arm (SE), HIGGS, 10 repeats (top) and 5 fresh repeats (bottom); bold is beyond 2 SE; the rule picks $\kappa = 1$ in every draw.

n	vs a4 (CV C)	vs free column, same rule	vs reference alone	$\kappa = 1$ picks
100	$-$**0.0160** (0.0041)	$-$0.0082 (0.0044)	**+0.0104** (0.0040)	10/10
150	$-$**0.0090** (0.0034)	$-$**0.0064** (0.0030)	**+0.0043** (0.0019)	10/10
250	$-$**0.0071** (0.0027)	$-$0.0022 (0.0018)	**+0.0027** (0.0011)	10/10
1,000	+0.0016 (0.0009)	+0.0005 (0.0005)	+0.0001 (0.0002)	10/10
Fresh draws, 5 repeats, after selection				
100	$-$0.0090 (0.0045)	$-$**0.0103** (0.0050)	**+0.0149** (0.0072)	5/5
150	$-$0.0121 (0.0068)	$-$0.0021 (0.0016)	**+0.0055** (0.0018)	5/5

C Origin

On a proprietary binary forecasting task with a strong external reference probability and a few hundred training examples, refitting a ridge logistic model with a fixed offset of 0.25 x the reference's log-odds lowered held-out log-loss on about a thousand examples from 0.320 to 0.306 relative to the same model without the offset (paired difference -0.014, moving-block bootstrap SE 0.0035). The fraction was not the best choice: the full offset ($\kappa = 1$) reached 0.283 (0.25 minus 1: +0.023, SE 0.005) and the reference log-odds as an ordinary, nearly unpenalised feature reached 0.287 (offset 0.25 minus feature: +0.018, SE 0.006), the full offset and the free feature being within 0.004 (SE 0.002) of each other.

D Reproduction

All code and results are at commit `6a6daad` of the `anchorlogit` repository. The patch is `patches/scikit-learn-offset.patch` (its full diff against scikit-learn 1.9.1); the benchmarks run with `bench/run_all.sh` and `bench/run_variants.sh`, and `python -m bench.summarize` prints every table. The fresh-draw check of the one-SE rule is `python paper/results/confirm_1se.py`. Numbers in this note come from `paper/results/make_numbers.py` and `make_extra.py`; recomputing the means and SEs from the stored per-row test log-odds differs from the result files by at most 2.3×10^{-9}. All runs used CPU only with 2 threads per process, 3.5 h in total (HIGGS 63 min, Elec2 38, Bank 31), Python 3.12.3, scikit-learn 1.9.1, statsmodels 0.15.0, glum 3.4.1, `anchorlogit` 0.1.0. On stock scikit-learn without the patch the package passes 288 tests and skips 10.